TSCG: The Transdisciplinary System Construction Game

- 1 TSCG: The Transdisciplinary System Construction Game
  - 1.1 A Proposal to Practice Systemic Modeling as a Construction Game
  - 1.2 Abstract
  - 1.3 1. Introduction
    - 1.3.1 1.1 The Problem of Disciplinary Silos and the “Desiloification” Hypothesis
    - 1.3.2 1.2 Origin: Twenty Years of Creative Meditation, Then an AI Interlocutor
    - 1.3.3 1.3 The Bicephalous Cyclops: Stereopsy of Knowledge
  - 1.4 2. The TSCG Architecture
    - 1.4.1 2.1 M3 — The Genesis Space
    - 1.4.2 2.2 M2 — The GenericConcepts Layer
    - 1.4.3 2.3 M1 — Domain Extensions
    - 1.4.4 2.4 M0 — Concrete Instances
    - 1.4.5 2.5 The Purity Principle: Why Domain Specialization is Deferred to M1 and M0
    - 1.4.6 2.6 TSCG Ontology Types
  - 1.5 3. The GenericConcepts Layer (M2)
    - 1.5.1 3.1 Tensor Products as a Concept Construction Language
    - 1.5.2 3.2 The Nine Families
    - 1.5.3 3.3 Bicephalous GenericConcepts
    - 1.5.4 3.4 GenericConceptCombos
    - 1.5.5 3.5 The v15.11.0 Additions: A Causal Chain for Irreversibility
    - 1.5.6 3.6 Mathematical Validation Status
  - 1.6 4. The Instance Corpus
    - 1.6.1 4.1 Defining the Poclet — A Neologism and Its Rationale
    - 1.6.2 4.2 The Validated Poclet Corpus (v15.11.0)
    - 1.6.3 4.3 Cross-Domain Pattern Resonance — A First Observation
    - 1.6.4 4.4 From Scripts to Gallery: The Ergonomic and Pedagogical Leap
  - 1.7 5. Noise Reduction Mechanisms
    - 1.7.1 5.1 GenericConceptCombo — Synergistic Combination at M2
    - 1.7.2 5.2 KnowledgeFieldConceptCombo — Domain Specialization at M1
  - 1.8 6. Validation Framework
    - 1.8.1 6.0 From System to Score: How Coefficients Are Assigned
    - 1.8.2 6.1 The Epistemic Gap δ as Primary Metric
    - 1.8.3 6.2 SpectralClasses — Interpreting the Gap
    - 1.8.4 6.3 Poclet Validation Checklist
    - 1.8.5 6.4 Cross-Poclet Validation — The Six-Domain Threshold
  - 1.9 7. Discussion
    - 1.9.1 7.1 “Is TSCG just another taxonomy?”
    - 1.9.2 7.2 “Why five ASFID dimensions? Why not four, or seven?”
    - 1.9.3 7.3 “Is the tensor formalism mathematically legitimate?”
    - 1.9.4 7.4 “Is 22 poclets a sufficient corpus?”
    - 1.9.5 7.5 “How can TSCG be falsified?”
    - 1.9.6 7.6 “Why does the scoring process rely on human judgment?”

- 1.9.7 7.7 On Assumed Arbitrariness
- 1.10 8. References
- 1.11 Supplementary Materials
- 1.12 Appendix B: Lessons from Human-AI Collaborative Research
  - 1.12.1 B.1 Context Window Management
  - 1.12.2 B.2 The Skills Pipeline System
  - 1.12.3 B.3 Hallucinations and Terminology Drift
  - 1.12.4 B.4 Output Review — A Differentiated Discipline
  - 1.12.5 B.5 From Scripts to Standalone HTML
  - 1.12.6 B.6 Overall Assessment
- 1.13 Appendix C: GenericConcepts Catalog (TSCG v15.11.0)
  - 1.13.1 Structural (24 concepts)
  - 1.13.2 Dynamic (18 concepts)
  - 1.13.3 Regulatory (8 concepts)
  - 1.13.4 Adaptive (7 concepts)
  - 1.13.5 Energetic (5 concepts)
  - 1.13.6 Informational (8 concepts)
  - 1.13.7 Ontological (9 concepts)
  - 1.13.8 Relational (8 concepts)
  - 1.13.9 Teleonomic (3 concepts)
  - 1.13.10 GenericConceptCombos (4 — defined in M2; instances in M1_CoreConcepts)
- 1.14 Appendix D: Instance Catalog (TSCG v15.11.0)
  - 1.14.1 D.1 Poclets (m3:Poclet) — see §4.2 for full table
  - 1.14.2 D.2 TransDisclets (m3:TransDisclet)
  - 1.14.3 D.3 TSCG Tools (m3:TscgTool)
  - 1.14.4 D.4 Systemic Frameworks (m3:SystemicFramework)
  - 1.14.5 D.5 Symbolic System Grammars (m3:SymbolicSystemGrammar)
- 1.15 Appendix E: Framework Maturity Stages
  - 1.15.1 E.1 — Stage 1: Transversal Systemic Principles
  - 1.15.2 E.2 — Stage 2: The Observer Problem
  - 1.15.3 E.3 — Stage 3: The Bicephalous Cyclops Metaphor
  - 1.15.4 E.4 — Stage 4: TSCG as a Meta-Map
  - 1.15.5 E.5 — Stage 5: The Poclet as Instrument
  - 1.15.6 E.6 — Stage 6: Pedagogy, Democratization, and Research Potential
  - 1.15.7 Summary Table

1 TSCG: The Transdisciplinary System Construction Game

1.1 A Proposal to Practice Systemic Modeling as a Construction Game

Authors: Echopraxium with the collaboration of Claude AI **Date**: March 2026 **Version**: 4.0 **Framework Version**: TSCG v15.11.0 **DOI (Prior Work)**: 10.5281/zenodo.18471860 **Repository**: https://github.com/Echopraxium/tscg **Live Demo**: https://echopraxium.github.io/tscg/ **License**: CC BY 4.0

1.2 Abstract

This paper presents the Transdisciplinary System Construction Game (TSCG) version 15.11.0, a modeling framework for analyzing and designing complex systems across disciplinary boundaries. Born from over twenty-five years of creative reflection on the existence of generic, recurring principles in most systems, and developed through sustained collaboration with Claude AI (Anthropic), the framework synthesizes systems theory, cybernetics, phenomenology, and semantic web technologies into a practical construction kit for systemic modeling.

TSCG proposes a **bicephalous architecture** grounded in an extended reading of Korzybski's map-territory distinction: the **Eagle Eye** (ASFID: Attractor, Structure, Flow, Information, Dynamics) measures the Territory as it is; the **Sphinx Eye** (REVOI: Representability, Evolvability, Verifiability, Observability, Interoperability) constructs the Map as it is known. Crucially, this duality is not a simple dichotomy but a bidirectional feedback loop: the Map shapes how the Territory is observed, and the Territory's resistance to inadequate Maps forces Map revision. The framework is formalized through tensor algebra and Hilbert space decomposition across a four-layer hierarchical ontology (M3→M2→M1→M0) implemented in JSON-LD semantic web standards.

The ontological core comprises **80 atomic GenericConcepts** organized into 9 families, each expressed as a tensor product of ASFID⊕REVOI dimensions (e.g., Synergy = D⊗I⊗R, Entropy = F⊗I⊗D, Homeostasis = A⊗S⊗F⊗V⊗E). Two noise-reduction mechanisms prevent ontological proliferation: **GenericConceptCombo** (synergistic tensor-product combinations achieving approximately 31% concept reduction at M2 level) and **KnowledgeFieldConceptCombo** (domain-specific parameterizable templates demonstrating up to 97% reduction in domain concept count). Validation is conducted through 22 **poclets** — minimal, complete, pedagogical instances of systems — spanning photography, Norse mythology, nuclear engineering, biology, electronics, music theory, blockchain consensus, and beyond.

A notable recent addition is the relaxation of a foundational M3 axiom: Flow is no longer required to be strictly positive ($F > 0$) but admits a zero state ($F \geq 0$), giving rise to a full spectrum from Stasis ($F = 0$, reversible) through latent potential (F_potential) to active morphism (F_active), and enabling the formalization of irreversibility through a causal chain: Dissipation → Entropy → Inertia → Absorbing State.

A poclet may be illustrated by its **standalone HTML simulation** — a zero-dependency interactive visualization accessible directly in any browser. Ten simulations are currently published, collectively forming the **TSCG Poclet Gallery** (https://echopraxium.github.io/tscg/). This represents a deliberate ergonomic and pedagogical shift: from environment-dependent Python scripts to universally accessible, link-shareable demonstrations.

In the spirit of a *Systemic Esperanto* — a shared vocabulary for recognizing structural kinship across disciplines — TSCG is submitted to the research community (researchers, engineers, students, and artists alike) as an open invitation to explore, challenge, extend, and ultimately contribute to the progressive identification of universal systemic patterns.

Keywords: Systems theory, Ontology engineering, Transdisciplinarity, Knowledge representation, Map-territory distinction, Cybernetics, Tensor products, Semantic web, ASFID, REVOI, Desiloification, Poclet, GenericConcept, JSON-LD

1.3 1. Introduction

1.3.1 1.1 The Problem of Disciplinary Silos and the "Desiloification" Hypothesis

Modern civilization confronts challenges — climate change, pandemic response, artificial intelligence governance, sustainable energy transitions — that are inherently systemic and transcend any single discipline. Yet our intellectual infrastructure remains fragmented. The nuclear engineer designing reactor safety systems, the photographer balancing the *Exposure Triangle*, the mythologist interpreting the Norse cosmological tree *Yggdrasil*, the nephrologist modeling the *RAAS* (Renin-Angiotensin-Aldosterone System), and the distributed systems researcher analyzing Nakamoto consensus in Bitcoin all grapple with structurally isomorphic challenges: maintaining equilibrium through balanced competing forces, managing trade-offs, designing for stability under perturbation, resisting entropy-driven degradation. Yet these practitioners have no shared language to recognize their kinship.

This fragmentation generates three compounding problems. First, **redundant reinvention**: *Negative feedback* (control theory), *Homeostasis* (biology), *Equilibration* (economics), and *Mean reversion* (finance) describe the same fundamental pattern, yet communities rarely cross-pollinate. Second, **integration barriers**: insights from hormonal cascade regulation in physiology go unrecognized as structural twins of neutron moderation in nuclear physics, or of proof-of-work difficulty adjustment in blockchain consensus. Third, **ontological proliferation**: without principled abstraction, knowledge systems explode with domain-specific concepts, becoming cognitively intractable.

The central hypothesis of TSCG — call it the *desiloification hypothesis* — is this: **most systems, whether natural or artificial, physical or abstract, share a set of generic, recurring, transdisciplinary principles that can be identified, formalized, and used as a shared vocabulary**. The project does not claim these principles are exhaustive or that they constitute a complete description of reality. It claims only that they are sufficiently common and structurally robust to serve as a practical construction kit for systemic modeling.

A corollary hypothesis — which TSCG's growing poclet corpus progressively supports — is that the *same tensor-product formula* can describe structurally homologous phenomena across entirely unrelated domains. Dissipation ($F \otimes D$) governs energy loss in thermodynamics, signal attenuation in electronics, and hash-rate decay in a declining proof-of-work network alike. This cross-domain resonance is not metaphor: it is the empirical claim that TSCG invites the research community to scrutinize and challenge.

TSCG does not emerge from a void. Several prior frameworks have pursued structurally related ambitions, and this work stands consciously on their shoulders. Stafford Beer's **Viable System Model** (VSM, 1972) demonstrated that the recursive structure of viable organizations — from single cells to corporations — could be described by a small set of cybernetic functions, independent of the domain. Genrich Altshuller's **TRIZ** (Theory of Inventive Problem Solving, developed 1946–1985) distilled forty inventive principles from the systematic analysis of hundreds of thousands of patents, revealing that inventive solutions recur across engineering domains: a strikingly early empirical validation of the desiloification hypothesis. Alfred Korzybski's **map-territory distinction** (1933) — "the map is not the territory" — provides TSCG's epistemological backbone, extended here into a formalized bidirectional feedback loop between representation and phenomenon. More broadly, TSCG inherits from the traditions of **General System Theory** (von Bertalanffy, 1968), **Cybernetics** (Wiener, 1948), **Category Theory** (Eilenberg & Mac Lane, 1945), and **Semantic Web** ontology engineering (Berners-Lee et al., 2001).

Where TSCG differs from these precedents is in its *combinatorial* approach: rather than proposing a fixed set of domain-specific patterns (as TRIZ does for engineering) or a single organizational archetype (as VSM does for viable systems), TSCG constructs a vocabulary of atomic transdisciplinary concepts — GenericConcepts — that combine through tensor products to generate domain-specific patterns on demand. The framework is deliberately open-ended and self-revising: a construction kit, not a finished cathedral. It assumes its share of arbitrariness — the choice of ASFID dimensions, the boundaries between GenericConcepts, the granularity of the M2 vocabulary — and submits these choices to the community for scrutiny, extension, and refutation.

This aspiration has an older precedent in linguistics. In 1887, Ludwig Lazarus Zamenhof proposed Esperanto as a constructed common language designed to transcend national linguistic barriers — not by replacing any natural language, but by offering a neutral shared medium. TSCG pursues an analogous ambition at the level of systemic knowledge: a **Systemic Esperanto**, a constructed common vocabulary for describing the structural behavior of systems across disciplines. Where Esperanto addressed the fragmentation of human communication — the *Babel problem* of natural languages — TSCG addresses the fragmentation of systemic knowledge: the *Babel problem* of disciplines. The goal is not *desiloification by erasure* (forcing all disciplines into a single mold) but *desiloification by translation*: providing a shared layer of abstraction through which practitioners from biology, engineering, economics, mythology, music theory, and blockchain research can recognize structural kinship without abandoning their own domain vocabulary.

TSCG makes no claim to be a new Theory of Everything. Its 80 GenericConcepts do not exhaust the structure of reality; its tensor formulas are not laws of nature. The vocabulary offered here involves deliberate, partially arbitrary choices — choices that are submitted, not imposed. The intended audience is broad: researchers seeking cross-disciplinary patterns, engineers designing complex systems, students building systemic intuition, and artists exploring structural analogies between creative and natural processes. The long-term scientific objective — modest in ambition, ambitious in scope — is the progressive, community-driven identification of **cross-disciplinary systemic patterns**: recurring structural signatures that appear, under different names and in different guises, across the full breadth of organized complexity.

1.3.2 1.2 Origin: Twenty Years of Creative Meditation, Then an AI Interlocutor

This framework did not emerge from an academic program. It grew from more than twenty-five years of informal, persistent reflection — what the author describes as “creative meditation” — on whether generic principles truly recur across most systems. The intuition was strong but remained largely qualitative: structural echoes between biological regulation and industrial control systems, between mythological cosmologies and network topologies, between photographic technique and feedback control.

The turning point came with the availability of Large Language Model (LLM) conversational agents. After exploratory conversations across several AI platforms, the author settled on **Claude AI** (Anthropic) as the primary collaborator, for a pragmatic reason: Claude supports persistent *Projects* with associated reference documents, enabling the accumulation and re-use of a corpus of framework specifications across conversations — essential for a project of this complexity and duration.

The author describes himself explicitly as a *non-mathematician*. This constraint shaped the collaboration model: rather than a tool executing instructions, Claude AI functions as an active intellectual partner, with a clear and deliberate division of labor.

1.3.2.1 Delegated tasks (Claude AI as executor)

Several categories of work were systematically delegated to Claude AI, freeing the author to focus on conceptual direction and validation:

Mathematical formulation. The tensor product formulas characterizing each GenericConcept, the Hilbert space decomposition of the ASFID⊕REVOI state vectors, and the epistemic gap δ calculations were formulated by Claude AI on the basis of the author's conceptual specifications. These formulations are acknowledged as *AI-assisted approximations* pending rigorous mathematical review. A planned future step is their formal encoding using **SHACL** (Shapes Constraint Language), which will provide machine-verifiable constraints on the ontological structures — moving from narrative mathematical claims to formally checkable specifications.

Documentary research per poclet. For each poclet under development, Claude AI conducted targeted literature research on the system being modeled — sourcing domain-specific descriptions, identifying key structural components, and synthesizing relevant cross-disciplinary analogies. This research delegation was essential for the rapid expansion of the poclet corpus across domains in which the author has no specialist expertise.

JSON-LD ontology encoding. The translation of conceptual models into well-formed JSON-LD files — respecting W3C standards, TSCG namespace conventions, and ontological consistency constraints — was delegated to Claude AI. The author reviewed and validated each generated file; encoding itself was AI-executed.

1.3.2.2 Collaborative tasks (Claude AI as active partner)

Other tasks required genuine intellectual collaboration rather than delegation — an iterative back-and-forth in which both parties contributed substantively:

GenericConcept identification and modeling (M2 layer). The identification of candidate GenericConcepts — particularly during poclet analysis, when a new system under scrutiny reveals structural patterns not yet captured in M2 — is a collaborative process. The author proposes candidates based on systemic intuition; Claude AI evaluates their tensor decomposition, checks consistency with existing M2 concepts, identifies potential overlaps or contradictions, and proposes formulations. The session of 23 March 2026, which produced five new atomic GenericConcepts (Entropy, Stase, Processor, Transducer, Coherence) and four new Combos, is a representative example of this collaborative mode.

Poclet simulation generation. Each poclet's standalone HTML simulation is generated collaboratively. Claude AI takes the initiative on the illustrative mechanisms — choosing visual representations, animation dynamics, and interactive controls that best convey the system's behavior. The author pilots the process from the perspective of ergonomics and pedagogical credibility: ensuring that the simulation is legible to a non-specialist, that the ASFID/REVOI scores are meaningfully visualized, and that the interactive controls reveal rather than obscure the system's dynamics. Several iterative rounds of feedback and refinement are typical for each simulation.

Workflow engineering via Skills pipelines. Most recently, the collaboration extended to the meta-level of the research process itself. Two structured AI *Skills* — reusable pipeline specifications encoding the author's accumulated methodology — have been formalized: a **Poclet Analysis Pipeline** (guiding the four-stage process of proposal, analysis, modeling, and simulation for each new poclet) and a **Research Article Pipeline** (guiding the staged drafting and updating of preprints, including the present paper). These Skills function as persistent, versioned methodological memory: they ensure that a new Claude AI session, initiated weeks or months after the previous one, recovers the full procedural context of the project without requiring the author to re-specify the workflow from scratch.

1.3.2.3 Continuity infrastructure

The collaboration is sustained across sessions by two complementary mechanisms. A structured **Smart Prompt** document (currently `TSCG_Smart_Prompt_v15_10_1.md` in the repository) summarizes all key architectural

decisions, naming conventions, and project state; it is re-injected at the start of each new AI session as a "working memory prosthesis." The **Skills** pipeline system complements this by encoding procedural knowledge — not what the framework is, but how to work with it — as reusable, versioned AI instructions.

The methodological lessons from this collaboration — including context window management, hallucination patterns, the necessity of systematic output review, and the productive tension between AI initiative and human validation — are documented in detail in Appendix B.

1.3.3 1.3 The Bicephalous Cyclops: Stereopsy of Knowledge

To understand TSCG's architecture, consider a creature from Mesopotamian mythology: **Anzû**, a divine entity depicted as a lion-headed eagle — simultaneously earthly and celestial, interpretive and calculative. In TSCG, Anzû serves as the foundational metaphor for the M3 layer: a single framework with two heads, each possessing one eye.

The critical detail is the *one eye per head*. In optics, a single eye produces **monocular vision**: a flat, two-dimensional projection with no depth perception. Two eyes, positioned at a slight angle to each other, produce **binocular vision**: the brain triangulates the two slightly different images into a three-dimensional depth map. The same principle governs epistemic architecture.

The Eagle Head scrutinizes the world as it *is* — measuring, instrumenting, quantifying phenomena. Its eye is the **Eagle Eye**, operating through the five ASFID dimensions: Attractor, Structure, Flow, Information, Dynamics. It asks of any system: what draws it toward equilibrium? what organizes it? what flows through it? what information governs it? what dynamics evolve it? The Eagle Eye produces a Territory state vector — a quantified description of the system's observable behavior.

The Sphinx Head contemplates the world as it is *represented* — modeling, theorizing, interpreting. Its eye is the **Sphinx Eye**, operating through the five REVOI dimensions: Representability, Evolvability, Verifiability, Observability, Interoperability. It asks of any model: can this system be adequately represented? can the model evolve as the system does? can its predictions be verified? can its behavior be observed? can it communicate with other models? The Sphinx Eye produces a Map state vector — a quantified assessment of the model's epistemic quality.

Neither head can function alone. The Eagle Head without the Sphinx Head produces raw measurements with no interpretive framework — data without meaning. The Sphinx Head without the Eagle Head produces elegant theories with no empirical grounding — meaning without data. Each head, alone, has monocular vision: a flat, depthless perspective. Together, they produce **stereoscopic epistemic depth**: the ability to perceive not only the system, but the *distance* between our model of it and the reality it represents.

This distance is formalized as the **epistemic gap δ**:

```
δ = ‖|Ω⟩ − F_implement(|M⟩)‖
```

where |Ω⟩ is the Territory state vector (ASFID scores), |M⟩ is the Map state vector (REVOI scores), and F_implement is the morphism that projects Map predictions onto Territory observations. A small δ indicates that the model adequately tracks the system; a large δ signals either model inadequacy or an under-observed Territory. TSCG classifies epistemic gaps into four spectral regions: **Coherent** (δ ∈ [0, 0.05)), **OnCriticalLine** ([0.05, 0.15)), **Liminal** ([0.15, 0.30)), and **Enigmatic** ([0.30, 1.0)).

Crucially, the relationship between Map and Territory is not one-directional. The Map shapes *how* the Territory is observed — the choice of measurement tools, the selection of relevant variables, the temporal and spatial scales attended to are all determined by the current theoretical framework. Simultaneously, the Territory *resists* inadequate

Maps: when predictions fail and δ grows, the Map must be revised. This bidirectional co-construction — formalized through the coupling matrix Σ linking the two state vectors — is TSCG's extension of Korzybski's original distinction. Where Korzybski established the *asymmetry* between Map and Territory, TSCG formalizes their *dynamic interdependence*.

The Bicephalous Cyclops is therefore not a static duality but a living feedback loop: the Eagle Eye measures, the Sphinx Eye models, the gap δ signals divergence, and both heads adjust — the Eagle by refining its instruments, the Sphinx by revising its representations. Systemic knowledge is the ongoing product of this loop, never its final resolution.

1.4 2. The TSCG Architecture

TSCG is organized as a **four-layer hierarchical ontology**, implemented in JSON-LD following W3C semantic web standards. The layers are ordered from the most abstract (M3) to the most concrete (M0), connected by three functors that progressively instantiate and specialize the framework's vocabulary:

```
M3  Genesis Space        — Bicephalous mathematical foundation (ASFID ⊕ REVOI, 10D)
 │  F_tensorization ↓
M2  GenericConcepts      — 80 atomic transdisciplinary concepts (9 families)
 │  F_instantiation ↓
M1  Domain Extensions    — Knowledge-field vocabularies (Biology, Chemistry, Music…)
 │  F_concretize ↓
M0  Poclets & Instances  — Minimal complete models of concrete systems
```

Each layer is a strict refinement of the one above: M2 concepts are tensor products of M3 dimensions; M1 concepts are instantiations of M2 concepts in specific domains; M0 instances are concrete realizations of M1 concepts in specific systems. No layer introduces vocabulary that belongs to a higher layer — a discipline enforced by the **Purity Principle** (§2.5).

1.4.1 2.1 M3 — The Genesis Space

The foundation of TSCG is a **10-dimensional Hilbert space** — the Genesis Space — formed by the orthogonal union of two 5-dimensional subspaces:

```
H_TSCG = H_ASFID ⊕ H_REVOI
```

The orthogonality of the two subspaces (⟨ASFID_i, REVOI_j⟩ ≈ 0 ∀i,j) has been verified, and the completeness theorem states that any system representation in TSCG belongs to the span of ASFID ∪ REVOI.

The ASFID subspace (Eagle Eye — Territory measurement) decomposes any system along five orthogonal dimensions: - **A — Attractor**: the goal state or equilibrium toward which the system tends - **S — Structure**: the stable organizational pattern of the system's components - **F — Flow**: the transfer of energy, matter, or information through the system - **I — Information**: the signals, codes, and representations governing behavior - **D — Dynamics**: the temporal evolution and state transitions of the system

The REVOI subspace (Sphinx Eye — Map construction) evaluates any model of a system along five orthogonal dimensions: - **R — Representability**: can the system be adequately modeled? - **E — Evolvability**: can the model evolve as the system changes? - **V — Verifiability**: can the model's predictions be tested against reality? - **O — Observability**: can the system's relevant states be perceived? - **I — Interoperability**: can the model communicate with other models?

The two subspaces are coupled by the **alignment operator Σ** (a 5×5 matrix of singular values encoding coupling strengths between ASFID and REVOI dimensions) and two feedback morphisms: - **Φ: ASFID → REVOI** (observation: Territory informs Map construction) - **Ψ: REVOI → ASFID** (interpretation: Map shapes how Territory is measured)

The iterative feedback loop $v_T^{(n+1)} = v_T^{(n)} + \alpha \cdot \Psi(\Phi(v_T^{(n)}))$ converges to a fixed point representing **epistemic equilibrium** — the state where the Map's predictions and the Territory's observations are maximally aligned.

1.4.1.1 2.1.1 The Flow Axiom and Its Recent Relaxation

In all versions of TSCG prior to v15.11.0, Flow was defined as strictly positive: $F > 0$. The analysis of the NakamotoConsensus poclet — grounded in the statistical physics framework of Ranaora & Yii (2026) — revealed this axiom was unnecessarily restrictive. The revised axiom reads: $\mathbf{F \geq 0}$. Flow = 0 is now a valid ground state, corresponding to the new GenericConcept **Stase** ($S \otimes A$) — a reversible suspension of flow in which the system's structure and attractor are preserved intact. This relaxation gives rise to a full **F-spectrum**:

```
F = 0              → Stase (ground state, reversible, D absent)
0 < F < F_crit     → F_potential (latent reserve, not yet active)
F = F_crit         → Potentialization threshold
F > F_crit         → F_active (effective morphism, transport in progress)
F → ∞              → Maximum dissipation (Prigogine far-from-equilibrium)
```

A further annotation: Flow is the only ASFID dimension that possesses an intrinsic dual nature as both entity and morphism — $F \in Mor(Cat_M3) \cap Ob(Cat_M3)$. Any M2 GenericConcept whose tensor formula contains F is therefore a natural candidate for entity/morphism duality.

1.4.2 2.2 M2 — The GenericConcepts Layer

M2 is the operational core of TSCG: a vocabulary of **80 atomic GenericConcepts** organized into 9 families (Structural, Dynamic, Regulatory, Adaptive, Energetic, Informational, Ontological, Teleonomic, Relational), each expressed as a tensor product of $ASFID \oplus REVOI$ dimensions.

GenericConcepts are constructed from M3 dimensions by the tensorization functor $F_\otimes: Cat_M3 \to Cat_M2$, from the tensor space $T_M2 = \otimes_{k=1}^{3} H_M3^{\otimes k}$, producing 155 possible tensor products. Of these positions, TSCG v15.11.0 occupies 80 with validated atomic GenericConcepts. The nine families with representative examples are listed in Appendix C.

Beyond atomic concepts, M2 defines two **combination classes**: - **GenericConceptCombo**: synergistic tensor-product combinations of existing GenericConcepts, producing emergent patterns not reducible to their components. Currently 4 validated Combos. - **KnowledgeFieldConceptCombo**: parameterizable templates combining GenericConcepts in domain-specific ways, absorbing domain specialization without polluting M2 (see §2.5 and §5).

1.4.3 2.3 M1 — Domain Extensions

M1 is the translation layer between M2's generic vocabulary and the specific languages of knowledge fields. It consists of:

- **M1_CoreConcepts**: root knowledge-field concepts, GenericConceptCombo instances, and KnowledgeFieldConceptCombo instances shared across domains

- **M1 extensions** (one per domain): `M1_Biology.jsonld`, `M1_Chemistry.jsonld`, `M1_Economics.jsonld`, `M1_Music.jsonld`, etc.

Each M1 extension maps domain-specific concepts onto M2 GenericConcepts via `instantiatesMetaconcept` properties. A researcher adding TSCG support for a new domain creates a new M1 extension without touching M3 or M2.[1]

> [1] *Existing upper ontologies (BFO, DOLCE, SUMO) pursue a complementary goal: modeling what entities are, rather than how systems behave. A future integration path via OWL/RDF mappings from M1 extensions to BFO/DOLCE is identified as a medium-term objective (Arp et al., 2015; Gangemi et al., 2002).*

1.4.4 2.4 M0 — Concrete Instances

M0 is the empirical layer: concrete models of specific systems, each implemented as a JSON-LD file referencing M1 and M2 concepts. The primary M0 ontology type is the **poclet** — defined and discussed in detail in §4. Additional M0 instance types are listed in §2.6.

1.4.5 2.5 The Purity Principle: Why Domain Specialization is Deferred to M1 and M0

A deliberate architectural decision governs the M2/M1 boundary: **domain-specific vocabulary is systematically deferred to M1 and M0**. M3 and M2 are designed to be maximally generic and domain-agnostic.

If domain concepts were allowed to infiltrate M2, the layer would rapidly balloon into an unmanageable collection of discipline-specific terms — a digital reproduction of the very disciplinary silos TSCG aims to bridge. *Combustion* belongs to chemistry; *Clonal Selection* belongs to immunology; *Hash Rate* belongs to blockchain engineering.

M2 therefore admits only concepts **empirically validated across at least six unrelated domains** without exception. A candidate GenericConcept failing this test is either relegated to M1 as a domain-specific pattern, or formalized as a **KnowledgeFieldConceptCombo** — a parameterizable template combining existing GenericConcepts in a domain-specific configuration.

The practical consequence: **M3 and M2 speak the language of all systems; M1 and M0 speak the language of each system.** The generic core remains stable; the domain vocabulary grows at the periphery.

1.4.6 2.6 TSCG Ontology Types

Type	Description
`m3:Poclet`	Minimal, complete, pedagogical model of a single-domain system
`m3:TransDisclet`	Model of a system that exists natively across ≥2 disciplines simultaneously
`m3:TscgTool`	Software tool operating reflexively on the TSCG framework itself
`m3:SystemicFramework`	External systemic framework modeled within TSCG (VSM, TRIZ…)
`m3:SymbolicSystemGrammar`	Symbolic system with generative grammar (I-Ching…)
`m3:Enigma`	System presenting an apparent impossibility within TSCG (planned)

The `m3:Enigma` type — currently planned — will formalize systems whose behavior challenges a framework axiom, serving as the primary vehicle for Popperian refutation within TSCG.

1.5 3. The GenericConcepts Layer (M2)

1.5.1 3.1 Tensor Products as a Concept Construction Language

M2 GenericConcepts are not defined by verbal description alone — each carries a **tensor formula** encoding its structural signature as a product of M3 dimensions. The tensor product ⊗ captures the idea that a GenericConcept requires the *simultaneous, non-separable* presence of multiple dimensions: Homeostasis (A⊗S⊗F⊗V⊗E) cannot be reduced to Attractor alone, nor to Structure alone, but emerges only from their joint activation.

Four morphism types relate GenericConcepts to each other: - **Inclusion ↪**: Homeostasis ↪ Regulation (specialization) - **Composition ∘**: Learning = Memory ∘ Adaptation - **Duality ^op**: Convergence^op = Divergence; Entropy^op = Negentropy - **Emergence ⇒**: M_A ⊗ M_B ⇒ M_C (GenericConceptCombo)

1.5.2 3.2 The Nine Families

GenericConcepts are organized into nine families. The complete listing with tensor formulas is in Appendix C; here we present each family with representative examples.

Structural (24 concepts): Patterns governing the organization of components in space and hierarchy. *Examples*: Network (S⊗I⊗F⊗D⊗Im), Hierarchy (A⊗S⊗V⊗R), Stase (S⊗A), Coherence (A⊗S⊗I⊗R⊗O).

Dynamic (18 concepts): Patterns governing temporal evolution and state transitions. *Examples*: Process (D⊗F⊗R), Bifurcation (A⊗D⊗S), FeedbackLoop (S⊗I⊗A⊗D⊗F⊗V⊗E), Processor (S⊗I⊗F⊗D).

Regulatory (8 concepts): Patterns governing system stability and control. *Examples*: Homeostasis (A⊗S⊗F⊗V⊗E), Regulation (A⊗S⊗F⊗V⊗R), Threshold (A⊗I⊗O).

Adaptive (7 concepts): Patterns governing learning and adjustment. *Examples*: Adaptation (I⊗F⊗D), Memory (D⊗F⊗D), Resilience (A⊗S⊗E).

Energetic (5 concepts): Patterns governing energy and flux transformation. *Examples*: Dissipation (F⊗D), Entropy (F⊗I⊗D), Transducer (F⊗S⊗I).

Informational (8 concepts): Patterns governing encoding, signaling, and representation. *Examples*: Signal (F⊗I⊗R⊗O), Code (S⊗I⊗V), Language (S⊗I⊗F⊗A⊗R).

Ontological (9 concepts): Patterns governing identity, existence, and systemic context. *Examples*: System (S⊗F⊗V), Observer (A⊗I⊗R), Context (O⊗R⊗Im⊗E).

Teleonomic (3 concepts): Patterns governing goal-directedness and purpose. *Examples*: Scope (A⊗I⊗S⊗V), Role (A⊗S⊗I⊗Im).

Relational (8 concepts): Patterns governing connections and interactions. *Examples*: Mediator (S⊗I⊗F⊗O), Link (S⊗I⊗F⊗V), Relation (S⊗I⊗A⊗Im).

1.5.3 3.3 Bicephalous GenericConcepts

A subset of GenericConcepts span *both* ASFID and REVOI dimensions in a single tensor formula. These **bicephalous GenericConcepts** capture phenomena that are simultaneously territorial (observable) and cartographic (representational) by nature. The most architecturally significant example is **Coherence** (A⊗S⊗I⊗R⊗O):

```
A  → alignment toward a common attractor     (Territory / Eagle Eye)
S  → globally continuous structure            (Territory / Eagle Eye)
I  → shared / compatible information          (Territory / Eagle Eye)
R  → Representability — coherence as modeled  (Map / Sphinx Eye)
O  → Observability — coherence as perceived   (Map / Sphinx Eye)
```

Coherence admits four regimes (quaternary polarity):

```
                    R⊗O strong        R⊗O weak
A⊗S⊗I strong  →  Verified truth     Misunderstood reality
A⊗S⊗I weak    →  Illusion/ideology  Total chaos
```

1.5.4 3.4 GenericConceptCombos

Beyond atomic GenericConcepts, M2 defines the **GenericConceptCombo** class for synergistic combinations producing emergent patterns not reducible to components. TSCG v15.11.0 defines four validated Combos:

Combo	Formula	Components	Polarity
Inertia	S⊗F⊗I⊗D	⊗⇒(Memory, Entropy)	Neutral
Potentialization	A⊗D⊗F	⊗⇒(Activation, Process)	Dual (+/-)
Absorbing State	S⊗A⊗F⊗I⊗D	⊗⇒(Stase, Entropy)	Neutral
Topological Defect	S⊗A⊗I⊗R⊗O	⊗⇒(Incoherence, Invariant)	Neutral

Potentialization resolves a long-standing tension: concepts like Resource, Storage, Hub, and Stateful — previously problematic because they implied F=0 — are now understood as systems in suspended Potentialization (F_potential mode).

Absorbing State is a Stase sealed irreversibly by Entropy production — a terminal state from which Potentialization is structurally impossible, in contrast to ordinary Stase which remains reversible by construction (D absent from S⊗A).

1.5.5 3.5 The v15.11.0 Additions: A Causal Chain for Irreversibility

The analysis of the NakamotoConsensus poclet produced the most significant single-session M2 expansion in the framework's history: **five new atomic GenericConcepts**, **four new Combos**, **one major concept revision**, and a **foundational M3 architectural amendment** (the F axiom relaxation, §2.1.1).

The five new atomic GenericConcepts:

Concept	Formula	Family	Polarity	Key insight
Processor	S⊗I⊗F⊗D	Dynamic	Neutral	Generic transformer; parent of Transducer and Dissipation
Transducer	F⊗S⊗I	Energetic	Neutral	Converts flux type; subClassOf Processor
Entropy	F⊗I⊗D	Energetic	Dual	Irreversibility measure; produced by Dissipation (Feynman)
Stase	S⊗A	Structural	Neutral	F=0 reversible ground state; D absent by construction
Coherence	A⊗S⊗I⊗R⊗O	Structural	Dual	Bicephalous; factual AND perceived alignment

Dissipation (F⊗D) was substantially revised: its tensor formula is unchanged but it is now typed as subClassOf Transducer with an explicit `m2:produces → Entropy` relation, reflecting Feynman's insight that *the equations of Dissipation are time-reversible; irreversibility emerges from the Entropy it produces*.

Together, these additions constitute a **causal chain for irreversibility**:

```
F_active   (available exergy)
    │
    ↓  Dissipation (F⊗D) — equations time-reversible (Feynman)
    │
F_degraded ──→  Entropy (F⊗I⊗D)   ← irreversibility emerges here
                    │
                    ↓  Memory (∫dτ)
                    │
                Inertia ⊗⇒(Memory, Entropy)
                    │
                    ↓  maximal → seals Stase
                    │
            Absorbing State ⊗⇒(Stase, Entropy)
```

The Energetic hierarchy is now:

```
Processor   (S⊗I⊗F⊗D)
    └── Transducer (F⊗S⊗I)   subClassOf Processor
            └── Dissipation (F⊗D)   subClassOf Transducer
                    └── produces → Entropy (F⊗I⊗D)
```

A prior inconsistency from v6.5.0 is also resolved: "Coherence^op = Entropy" was imprecise. The correction: Coherence^op = Incoherence; Entropy^op = Negentropy. The relationship Incoherence → [Dissipation] → Entropy is a causal morphism, not categorical duality.

1.5.6 3.6 Mathematical Validation Status

The tensor formulas in this section were developed collaboratively with Claude AI and carry the ontology's own assessment: *preliminary, AI-validated, pending expert review*. The planned path toward formal validation is SHACL

encoding — expressing tensor-product constraints as machine-verifiable SHACL shapes — identified as a priority for a future release.

1.6 4. The Instance Corpus

1.6.1 4.1 Defining the Poclet — A Neologism and Its Rationale

The primary empirical instrument of TSCG is the **poclet** — a term coined for this framework, requiring explicit definition before use.

The word is a portmanteau of *Proof of Concept* and the diminutive suffix *-let* (as in *booklet*, *droplet*, *wavelet*): a small, self-contained proof of concept. More precisely, a poclet is a **minimal, complete, pedagogical model of a specific system**, formalized as a JSON-LD ontology file (M0 layer) that instantiates M1 and M2 concepts against a concrete system chosen from any domain.

Three properties are jointly required:

Minimality: the poclet models *only* what is necessary to capture the system's essential structural behavior. No component, GenericConcept, or score is included without justification.

Completeness: the poclet must be sufficient to characterize the system along all active ASFID dimensions. Full 5/5 coverage is the target, with partial coverage acceptable only when dimensions are genuinely absent.

Pedagogical intentionality: a poclet is designed to *teach*. The choice of system, the selection of illustrative GenericConcepts, the accompanying README, and the interactive simulation are oriented toward making the system's structural behavior legible to a domain-unfamiliar reader.

1.6.2 4.2 The Validated Poclet Corpus (v15.11.0)

TSCG v15.11.0 comprises **22 validated poclets** spanning 10+ domains.

ID	Poclet	Domain	ASFID	Key GenericConcepts	Sim.
M0-01	FireTriangle	Chemistry	5/5	Balance, Constraint, SelfSustainingReaction	✓
M0-02	FourStrokeEngine	Engineering	5/5	Cycle, Regulation, Process	✓
M0-03	CellSignalingModes	Biology	5/5	Network, Signal, Mediator	—
M0-04	Yggdrasil	Norse Mythology	5/5	Network, Bifurcation, Invariant	—
M0-05	BloodPressureControl	Biology/Medicine	5/5	Homeostasis, Regulation, FeedbackLoop	—
M0-06	ColorSynthesis	Optics	5/5	Polarity, Trade-off, Emergence	✓
M0-07	ExposureTriangle	Photography	5/5	Balance, Trade-off, Regulation	✓
M0-08	NuclearReactorTypology	Energy Engineering	5/5	Regulation, Constraint, Threshold	—
M0-09	Kidneys	Biology/Medicine	5/5	Homeostasis, Filtration, Threshold	—
M0-10	Transistor	Electronics	5/5	Regulation, Amplification, Threshold	✓
M0-11	RAAS	Biology/Medicine	5/5	Cascade, Amplification, Homeostasis	—
M0-12	ButterflyMetamorphosis	Biology	5/5	Bifurcation, Emergence, Stase	—
M0-13	AdaptiveImmuneResponse	Biology/Immunology	5/5	Memory, Network, Regulation	—
M0-14	ComplexChemicalSynapse	Neurobiology	5/5	Synergy, Process, Homeostasis	—
M0-15	TrophicPyramid	Ecology	5/5	Hierarchy, Flow, Constraint	—
M0-16	TvTestPattern	Broadcast Engineering	4/5	Code, Signal, Representation	—
M0-17	KindlebergerMinsky	Economics	5/5	Cycle, Bifurcation, Cascade	—
M0-18	PhaseTransition	Physics/Thermodynamics	5/5	Bifurcation, Threshold, Emergence	✓
M0-19	VCO	Electronics/Signal Processing	5/5	Oscillation, FeedbackLoop, Trajectory	✓
M0-20	CounterPoint	Music Theory	5/5	Constraint, Balance, Emergence	✓

ID	Poclet	Domain	ASFID	Key GenericConcepts	Sim.
M0-21	MtgColorWheel	Game Design/Semiotics	5/5	Balance, Polarity, Identity	✓
M0-22	NakamotoConsensus	Distributed Systems/Physics	5/5	Dissipation, Entropy, Inertia	✓

✓ = standalone HTML simulation available. — = simulation in progress.

NakamotoConsensus (M0-22) is the most recent and architecturally consequential addition: its analysis directly motivated the v15.11.0 M2 expansions (§3.5).

1.6.3 4.3 Cross-Domain Pattern Resonance — A First Observation

Across the 22-poclet corpus, certain GenericConcept combinations recur with striking regularity across unrelated domains. Three patterns stand out:

Homeostasis-Regulation-FeedbackLoop appears in BloodPressureControl, RAAS, Kidneys, NakamotoConsensus (difficulty adjustment), and NuclearReactorTypology (reactivity control). The same tensor signature governs hormonal cascades, nuclear criticality management, and blockchain consensus alike.

Bifurcation-Emergence-Threshold appears in ButterflyMetamorphosis, PhaseTransition, KindlebergerMinsky, and Yggdrasil — linking biological metamorphosis, phase changes, financial crises, and mythological catastrophe.

Dissipation-Entropy-Inertia — formalized as a connected causal chain in §3.5 — appears in NakamotoConsensus, FourStrokeEngine, and RAAS. Its formalization as a GenericConceptCombo chain rather than three isolated concepts represents the kind of structural discovery the poclet methodology is designed to produce.

These observations are preliminary — the corpus is too small for strong empirical claims. They are offered as *invitations to scrutiny*.

1.6.4 4.4 From Scripts to Gallery: The Ergonomic and Pedagogical Leap

1.6.4.1 4.4.1 The Standalone HTML Solution

Ten poclet simulations are currently implemented as **standalone HTML files** — single-file, zero-dependency applications running in any modern browser. The technical architecture: all data embedded as JavaScript literals (no `fetch()` calls), visualization libraries from cdnjs.cloudflare.com CDN, a Content Security Policy header ensuring Electron compatibility, the complete application in a single `.html` file.

All simulations follow a canonical layout (established by the FireTriangle simulation): CSS Grid with header, animation canvas, resizable sidebar (tabs: Description | Concepts | Synergy | Scores), and JS-based tooltips using `getBoundingClientRect()`.

1.6.4.2 4.4.2 The TSCG Poclet Gallery

These simulations are published at **https://echopraxium.github.io/tscg/** as a navigable **Poclet Gallery**, with the index generated automatically by `generate_index.js`. A conference presenter can demonstrate a poclet by sharing a URL;

a student can explore the full corpus from a smartphone; a researcher can link directly to a simulation from a citation. The framework's empirical claims are live, interactive demonstrations — one click away.

1.7 5. Noise Reduction Mechanisms

A framework whose ambition is transdisciplinary coverage faces an inherent tension: richer domain coverage generates pressure to introduce domain-specific concepts. TSCG employs two complementary noise-reduction mechanisms.

1.7.1 5.1 GenericConceptCombo — Synergistic Combination at M2

Rather than introducing a new atomic GenericConcept for every emergent pattern, TSCG provides the **GenericConceptCombo** class — a formal mechanism for expressing synergistic combinations of existing GenericConcepts that produce emergent patterns not reducible to their components. The notation ⊗⇒(M_A, M_B, …, M_n) denotes an N-ary emergent combination.

Architecture: The GenericConceptCombo *class* is defined in M2; Combo *instances* are hosted in M1_CoreConcepts.jsonld. This preserves M2 purity while making Combo instances available across all M1 domain extensions.

The nine currently validated GenericConceptCombos in M1_CoreConcepts:

Combo	Formula	Components
Cascade	S⊗I⊗A⊗D⊗F	⊗⇒(Process, Step, Trajectory)
Oscillator	A⊗S⊗F⊗D	⊗⇒(FeedbackLoop, Trajectory)
Propagation	S⊗I⊗A⊗D⊗F⊗V⊗Im	⊗⇒(Cascade, Duplication, Network)
Narration	S⊗I⊗D⊗F⊗V⊗R⊗A⊗Im	⊗⇒(Transformation, Representation, Relation)
ButterflyEffect	A⊗S⊗I⊗D⊗F	⊗⇒(Amplification, Bifurcation)
LALI	A⊗S⊗I⊗F⊗D	⊗⇒(Activation, Inhibition, Network)
StratifiedDissipation	S⊗F⊗D⊗I	⊗⇒(Layer, Dissipation)
CascadeAmplification	S⊗I⊗A⊗D⊗F⊗R	⊗⇒(Cascade, Amplification)
Inertia	S⊗F⊗I⊗D	⊗⇒(Memory, Entropy)

The Combo mechanism achieves an estimated **~31% reduction** in M2 concept count relative to a naive enumeration of all patterns in the 22-poclet corpus.

1.7.2 5.2 KnowledgeFieldConceptCombo — Domain Specialization at M1

A **KnowledgeFieldConceptCombo** is a parameterizable template combining existing GenericConcepts in a domain-specific configuration, absorbing a family of related domain concepts into a single reusable structure.

The NuclearReactorTypology demonstration: A naive approach would have introduced six M1 classes — one per reactor type (PWR, BWR, PHWR, RBMK, HTGR, FBR). Instead, a single KnowledgeFieldConceptCombo template

captures the structural pattern common to all reactor types, with six configurations parameterizing domain-specific variations (moderator, coolant, criticality regime, containment). This achieves a **~97% reduction** in domain concept count for this family.

Architecture: The KnowledgeFieldConceptCombo class is defined in M2 as a subclass of GenericConceptCombo; instances are hosted in M1_CoreConcepts or domain M1 extension files. Each instance specifies the GenericConcept parents (via `m2:combinesConcepts`), the knowledge field (via `m1:knowledgeField`), and valid parameter values (via `m2:ValueSpace` attributes).

Together, the two mechanisms implement TSCG's noise-reduction strategy at different levels: GenericConceptCombo prevents M2 bloat; KnowledgeFieldConceptCombo prevents M1 bloat. The framework can grow in domain coverage without growing proportionally in ontological complexity.

1.8 6. Validation Framework

1.8.1 6.0 From System to Score: How Coefficients Are Assigned

Before the epistemic gap δ can be computed, two sets of coefficients must be determined: the ASFID scores (Territory state vector |Ω⟩) and the REVOI scores (Map state vector |M⟩). The process is currently **semi-formal**: grounded in explicit criteria and documented systematically, but not yet fully automated.

1.8.1.1 ASFID Scoring — Measuring the Territory

Each ASFID dimension is scored on [0, 1] by the analyst using the following operational questions:

Dimension	Operational question	Score = 0	Score = 1
A Attractor	Does the system tend toward a definable equilibrium or stable state?	No identifiable attractor	Strong, well-defined attractor
S Structure	Does the system have a stable, identifiable organizational pattern?	Amorphous	Rich, hierarchical structure
F Flow	Does the system involve transfer of energy, matter, or information?	F = 0 (Stase)	Maximum active flux
I Information	Does the system process, encode, or transmit governing signals?	No information processing	Dense, multi-channel information
D Dynamics	Does the system evolve over time through state transitions?	Static, time-invariant	Rich temporal evolution

1.8.1.2 REVOI Scoring — Evaluating the Map

Dimension	Operational question	Score = 0	Score = 1
R Representability	Can this system be adequately represented as a formal model?	Fundamentally unmodelable	Completely and precisely modelable
E Evolvability	Can the model update as the system changes without structural revision?	Rigid, brittle	Fully parametric, extensible
V Verifiability	Can the model's predictions be empirically tested?	No testable predictions	All predictions falsifiable and tested
O Observability	Can the system's relevant states be perceived and measured?	Fully opaque	All states directly measurable
I Interoperability	Can this model communicate with and integrate into other models?	Isolated	Fully interoperable

REVOI scores are inherently **observer-relative**: the same system may receive different REVOI scores from a biologist and an engineer. This relativity is not a defect — it is a feature of the Sphinx Eye architecture (§1.3, Appendix E §E.2).

1.8.1.3 Worked Example — FireTriangle Poclet

The FireTriangle poclet illustrates the scoring process concretely.

ASFID scoring with justification:

Dim	Score	Justification
A	0.90	Strong attractor: combustion tends toward stoichiometric equilibrium. Slightly below 1.0 because extinction is always possible via removal of any vertex.
S	0.85	Clear triangular structure: three components with defined roles (fuel = substrate, oxidizer = reactant, heat = catalyst/product).
F	0.95	Dominant flow: energy release, oxygen consumption, combustion products. Flow is the defining characteristic of active combustion.
I	0.70	Moderate information: threshold conditions and feedback signals. Less information-dense than biological signaling systems.
D	0.80	Clear dynamics: ignition, sustained combustion, extinction. Relatively simple compared to multi-stage systems.

ASFID state vector: $|\Omega\rangle = 0.90|A\rangle + 0.85|S\rangle + 0.95|F\rangle + 0.70|I\rangle + 0.80|D\rangle$ **Norm**: $\||\Omega\rangle\| \approx 1.88$

REVOI scoring with justification:

Dim	Score	Justification
R	0.95	Canonical centuries-old representation with universal recognition.
E	0.75	Extends naturally to the fire tetrahedron without structural revision.
V	0.90	All predictions experimentally testable and routinely tested.
O	0.85	All three components directly measurable.
I	0.80	Interfaces naturally with thermodynamics, safety engineering, ecology.

Epistemic gap computation (δ_1):

Applying the diagonal Σ pairing (R↔I, E↔D, V↔A, O↔F, Im↔S):

```
F_implement(|M⟩) → 0.85|A⟩ + 0.80|S⟩ + 0.85|F⟩ + 0.95|I⟩ + 0.75|D⟩
                     ↑V          ↑Im         ↑O          ↑R         ↑E

Difference: 0.05|A⟩ + 0.05|S⟩ + 0.10|F⟩ − 0.25|I⟩ + 0.05|D⟩
‖difference‖ ≈ 0.277
δ₁ = 0.277 / 1.88 ≈ 0.147
```

SpectralClass: OnCriticalLine — driven primarily by the I dimension: the Map over-estimates informational richness relative to the Territory measurement. This identifies a concrete improvement direction for the FireTriangle scoring.

1.8.1.4 Calibration by Analogy

When scoring a new poclet, the analyst compares explicitly with existing poclets. Example — NakamotoConsensus vs FireTriangle:

Dim	FireTriangle	NakamotoConsensus	Rationale
A	0.90	0.85	Nakamoto's difficulty adjustment slightly less precise than stoichiometry
S	0.85	0.90	Nakamoto has richer structure (nodes, miners, mempool, chain)
F	0.95	0.95	Both dominated by active flow
I	0.70	0.95	Nakamoto is information-dense: Merkle trees, nonce search, difficulty target
D	0.80	0.90	Nakamoto exhibits richer dynamics: 2016-block adjustment, fork resolution

All scoring decisions and calibration comparisons are documented in each poclet's JSON-LD file under `m0:scoringJustification`, providing a persistent, version-controlled audit trail. The scoring process is **human-in-the-loop and semi-formal**: reproducible by a trained analyst following these guidelines, but not yet automated or statistically validated for inter-rater reliability.

The roadmap toward full formalization: (1) SHACL constraints on score ranges (planned v16.x); (2) a calibration corpus of reference poclets with consensus scores; (3) AI-assisted first-pass scoring via TscgPocletMiner.

1.8.2 6.1 The Epistemic Gap δ as Primary Metric

Territory state vector (Eagle Eye / ASFID):

```
|Ω⟩ = αₐ|A⟩ + αₛ|S⟩ + αf|F⟩ + αᵢ|I⟩ + αd|D⟩
```

Map state vector (Sphinx Eye / REVOI):

```
|M⟩ = βᵣ|R⟩ + βₑ|E⟩ + βᵥ|V⟩ + βₒ|O⟩ + βᵢ|I⟩
```

Three formulations of the epistemic gap, in increasing mathematical sophistication:

δ_0 — Scalar gap (rapid assessment):

```
δ₀ = |‖|Ω⟩‖ − ‖|M⟩‖| / max(‖|Ω⟩‖, ‖|M⟩‖)
```

δ_1 — Normalized vectorial gap (recommended for poclet scoring):

```
δ₁ = ‖|Ω⟩ − F_implement(|M⟩)‖ / ‖|Ω⟩‖
```

δ_2 — Riemannian gap (advanced geometric analysis):

```
δ₂ = d_g(|Ω⟩, |M⟩)
```

where d_g is the geodesic distance on the manifold defined by the coupling matrix Σ. Reserved for post-convergence geometric characterization.

1.8.3 6.2 SpectralClasses — Interpreting the Gap

SpectralClass	δ_1 Range	Interpretation
Coherent	[0.00, 0.05)	Map and Territory tightly aligned. High confidence in GenericConcept assignments.
OnCriticalLine	[0.05, 0.15)	Acceptable alignment with identifiable tension. Targeted refinement indicated.
Liminal	[0.15, 0.30)	Significant gap. Systematic review of ASFID/REVOI coefficients required.
Enigmatic	[0.30, 1.00]	Fundamental misalignment — candidate `m3:Enigma` or scope revision required.

An Enigmatic poclet is not a failed model — it is the most scientifically valuable outcome, signaling a potential framework refutation or M2 vocabulary gap.

1.8.4 6.3 Poclet Validation Checklist

Each poclet undergoes validation against four criteria:

ASFID completeness: All five dimensions explicitly scored and justified. Unscored dimensions must be argued as absent, not silently omitted.

GenericConcept coverage: Each mobilized GenericConcept plays a demonstrable, non-redundant structural role.

Minimality: Every component is necessary. Can any be removed without losing explanatory power?

Emergence: Does the combination of GenericConcepts produce insights not present in any single concept? Does the poclet reveal a structural analogy to at least one other domain?

1.8.5 6.4 Cross-Poclet Validation — The Six-Domain Threshold

A candidate GenericConcept must demonstrate applicability across **at least six unrelated knowledge fields** before M2 admission. The NakamotoConsensus poclet provided the sixth-domain validation for Dissipation and Entropy — confirming their M2 status by demonstrating structural relevance in distributed consensus alongside thermodynamics, biology, electronics, photography, and engineering.

1.9 7. Discussion

This section addresses the framework from the perspective of a skeptical reader. Each subsection names an objection directly and responds as honestly as possible — identifying which objections are fully answered, partially answered, and genuinely open.

1.9.1 7.1 “Is TSCG just another taxonomy?”

Objection: *You have simply collected existing systems theory concepts, given them new names, and arranged them in a table.*

Response: The objection is partially correct. Many individual GenericConcepts — Homeostasis, Regulation, Emergence, Network — are not new. TSCG does not claim to have invented them. What TSCG adds is threefold: a **formal construction language** (tensor formulas that make relationships computationally tractable); a **combinatorial mechanism** (GenericConceptCombos generating emergent concepts not in any existing taxonomy); and a **validation discipline** (the six-domain threshold and epistemic gap δ transform a collection of concepts into an empirical research program). The objection would be fatal if TSCG merely relabeled existing concepts. It is less compelling against a framework that assigns formal structure to those concepts and provides a methodology for testing their generality.

1.9.2 7.2 “Why five ASFID dimensions? Why not four, or seven?”

Objection: *The choice of exactly five ASFID dimensions is arbitrary.*

Response: The choice is **empirically motivated but not logically necessary**. The five dimensions emerged from twenty-five years of iterative reflection and validation. Two observations support their current form: across 22 poclets, no system has been encountered requiring a dimension not reducible to ASFID; and no two ASFID dimensions have been found redundant. The objection nevertheless identifies a genuine open question: are the ASFID dimensions the optimal basis? TSCG’s response is Popperian: submit the current choice to the community and let comparative explanatory power be assessed against the poclet corpus.

1.9.3 7.3 “Is the tensor formalism mathematically legitimate?”

Objection: *You use $\otimes$ in a way that does not correspond to standard mathematical usage.*

Response: This objection is **fully acknowledged** — and documented in the M2 ontology itself, which carries an explicit validation status of *“preliminary, AI-assisted, pending expert review.”* The tensor product notation is used analogically: capturing compositionality and dimensional non-separability in an intuitive notation, not a rigorously defined algebraic operation. The planned path toward legitimacy is SHACL encoding followed by expert review. Until then, tensor formulas should be read as **structured mnemonics** — precise enough to distinguish concepts and support pattern matching, but not yet proven to satisfy tensor algebra axioms in the strict sense.

1.9.4 7.4 “Is 22 poclets a sufficient corpus?”

Objection: *Twenty-two poclets is a small sample. Cross-domain pattern claims are statistically insignificant.*

Response: The objection is **correct as a statistical claim** and should temper confidence in the §4.3 observations, which are explicitly preliminary. Three contextualizing points: the corpus is **systematically diverse** (domains with no shared theoretical tradition); it is **actively growing** via the poclet pipeline; and the framework’s value is **not solely empirical** — even if cross-domain pattern claims remain contested, TSCG provides a structured vocabulary whose utility can be assessed independently of whether its patterns are universal laws or useful heuristics.

1.9.5 7.5 “How can TSCG be falsified?”

Objection: *A framework applicable to any system — from Norse mythology to blockchain consensus — is not falsifiable.*

Response: The objection identifies a genuine tension. TSCG is partly a language — and languages are not falsified, they are abandoned when less useful. But TSCG also makes substantive falsifiable claims:

Claim 1 — ASFID sufficiency: Every system of organized complexity can be characterized using the five ASFID dimensions. Falsification: a system where all five ASFID scores are identically zero despite exhibiting organized behavior.

Claim 2 — GenericConcept generality: Each M2 GenericConcept is structurally relevant in at least six unrelated domains. Falsification: a concept applicable in fewer than six domains upon careful analysis.

Claim 3 — Tensor formula consistency: Two poclets sharing a GenericConcept should exhibit structurally homologous behavior along its tensor dimensions. Falsification: contradictory structural signatures from the same concept.

The `m3:Enigma` type is specifically designed to document systems resisting TSCG modeling. An accumulation of Enigma instances in a specific dimensional region would constitute evidence against the framework's claims in that region.

1.9.6 7.6 "Why does the scoring process rely on human judgment?"

Objection: *ASFID and REVOI scores are assigned by a single analyst. This is subjective and not reproducible.*

Response: The objection is **valid and important**. The scoring process (§6.0) is currently semi-formal: grounded in explicit operational criteria, calibrated against existing poclets, and documented in each poclet's JSON-LD under `m0:scoringJustification` — but not yet validated for inter-rater reliability. TSCG does not claim otherwise. The roadmap toward fuller reproducibility is explicit: SHACL constraints on score ranges, a calibration corpus with consensus scores from multiple analysts, and AI-assisted first-pass scoring. Until these steps are completed, scores should be read as **documented expert estimates** whose derivation is traceable, not as objective measurements.

1.9.7 7.7 On Assumed Arbitrariness

A final meta-observation: **you may be right**. TSCG is twenty-five years of creative intuition formalized with AI assistance, validated against a small corpus, and submitted — not imposed — to a community that has not yet had the opportunity to engage with it systematically.

The choices embedded in this framework — the five ASFID dimensions, the nine M2 families, the minimality criterion, the six-domain threshold — are the author's best current answers to hard questions about building a transdisciplinary vocabulary. They assume their share of arbitrariness. Alternative choices may produce better frameworks.

What TSCG offers the skeptical reader is not certainty but **a concrete proposal to disagree with**. A framework sufficiently specific to be wrong is more scientifically valuable than one too vague to be tested. The invitation is to engage at the level of the specific claim — a tensor formula, a threshold, a SpectralClass boundary — and propose a better one.

1.10 8. References

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1.11 Supplementary Materials

Full documentation, interactive simulations, and extended appendices are available in the TSCG repository:

Repository: https://github.com/Echopraxium/tscg | **Live Poclet Gallery**: https://echopraxium.github.io/tscg/ | **Prior preprint (v3.0)**: https://doi.org/10.5281/zenodo.18471860

1.12 Appendix B: Lessons from Human-AI Collaborative Research

This appendix documents the methodological experience of developing TSCG through sustained collaboration with Claude AI.

1.12.1 B.1 Context Window Management

Claude AI Pro provides a context window of approximately 200,000 tokens — substantial, but insufficient for a project of this scope. A new conversation session starts without any memory of previous sessions, requiring explicit re-injection of project context before work can resume.

Mitigation: A compact **Smart Prompt** document (`TSCG_Smart_Prompt_v15_10_1.md`) summarizes all key architectural decisions, naming conventions, and project state. It is re-injected at the start of each new session as a working memory prosthesis.

1.12.2 B.2 The Skills Pipeline System

Beyond the Smart Prompt — which encodes *what* the framework is — a **Skills** system encodes *how* to work with it. Two pipeline Skills have been formalized: the Poclet Analysis Pipeline (four stages: proposal → analysis → modeling → simulation) and the Research Article Pipeline (five stages: audit → planning → drafting → revision → finalization). Each Skill is a versioned Markdown document encoding methodology, decision points, and synchronization checkpoints.

1.12.3 B.3 Hallucinations and Terminology Drift

The most persistent hallucination pattern: reversion to deprecated conventions when project context was thin — substituting *ORIVE* for *REVOI*, or describing R as *Reproducibility* rather than *Representability*.

Mitigation: A terminology watchdog — a lookup table of forbidden substitutions — embedded in the Smart Prompt and activated at the start of every session.

1.12.4 B.4 Output Review — A Differentiated Discipline

Not all AI-generated outputs receive the same level of human review before being committed to the repository. The author's level of engagement varies systematically by output type, reflecting the division of competencies documented in §1.2.

Mathematical formulas (tensor products, Hilbert space decompositions, epistemic gap calculations) are reviewed for *structural plausibility* and *internal consistency* — not for mathematical correctness in the strict sense. The author is, by his own description, a non-mathematician: the formalization of ASFID⊕REVOI dimensions, the construction of the tensor space T_M2, and the δ_1 formula were proposed by Claude AI and accepted on the basis of conceptual coherence with the framework's intentions. Independent mathematical expert review is explicitly identified as a pending step (§3.6, §7.3).

JSON-LD ontology encoding is similarly delegated: the author specifies the conceptual content (which concepts to add, what their properties and relationships should be), and Claude AI translates this into well-formed JSON-LD respecting W3C standards and TSCG namespace conventions. The author reviews the *conceptual* content of each generated file — verifying that the model reflects his intentions — not the syntactic correctness of the JSON-LD itself, which he trusts to the AI and to schema validation tooling.

GenericConcept identification and modeling is the one area where the review dynamic is genuinely bilateral. Candidate concepts are debated: the author proposes on the basis of systemic intuition; Claude AI evaluates tensor

decomposition and consistency; both parties can object. The resulting concepts are *co-produced*, not delegated. This "pair modeling" mode — analogous to pair programming — is what distinguishes M2 work from the other output categories.

Simulation code and interactive HTML is reviewed primarily for ergonomics and pedagogical credibility. The author iterates on visual clarity, tab structure, and interactive controls; he does not review the JavaScript or BabylonJS mechanics in detail.

In practice, this differentiated discipline has caught several classes of errors: UTF-8 corruption of mathematical symbols (caught during conceptual review of JSON-LD content), ontological inconsistencies such as tensor formula collisions (caught during pair modeling sessions), and simulation ergonomic issues (caught during interactive testing). It has *not* provided independent mathematical validation — a limitation honestly documented throughout this paper.

1.12.5 B.5 From Scripts to Standalone HTML

The migration from Python/Pygame to standalone HTML files eliminated the installation barrier entirely. Claude AI takes initiative on illustrative mechanisms; the author steers ergonomics and pedagogical credibility through iterative feedback rounds.

1.12.6 B.6 Overall Assessment

Productive human-AI partnership requires significant human effort in knowledge management. The AI is a capable, tireless, and creative collaborator — but the Smart Prompt, Skills pipelines, systematic review discipline, and the author's domain intuition are all indispensable. The human supplies memory, judgment, and long-term vision; the AI supplies formalization, breadth, and tireless iteration.

1.13 Appendix C: GenericConcepts Catalog (TSCG v15.11.0)

This appendix lists all 80 atomic GenericConcepts in TSCG v15.11.0, organized by family. Polarity: N = Neutral, D = Dual, H = Hybrid (bicephalous ASFID+REVOI). New concepts added in v15.11.0 are marked †. Authoritative source: `M2_GenericConcepts.jsonld` (https://github.com/Echopraxium/tscg)

1.13.1 Structural (24 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Capacity	S⊗I⊗A⊗E⊗O	N	Modularity	S⊗I⊗Im	N
Channel	S⊗I⊗F⊗E	N	Network	S⊗I⊗F⊗D⊗Im	N
Cluster	A⊗I⊗S⊗D⊗O	N	Node	S⊗I⊗F⊗Im	N
Component	S⊗I⊗F⊗R	N	Path	S⊗I⊗D⊗O	N
Composition	S⊗I⊗A	D	Polarity	A⊗I⊗S⊗R	N
Hierarchy	A⊗S⊗V⊗R	N	Segmentation	S⊗I⊗D⊗R	N
Hub	S⊗F⊗E	N	Symmetry	S	N
Identity	S⊗I⊗A⊗V⊗E	N	Topology	S⊗I⊗A⊗V	N
Imbrication	S⊗S	N	Workflow	D⊗F⊗S⊗V	N
Interoperability	S⊗I⊗F⊗V⊗E	N	Layer	S⊗I⊗A⊗R	N
Invariant	A⊗S⊗V	N	Space	S⊗I⊗D⊗E	N
Stase †	S⊗A	N	**Coherence** †	A⊗S⊗I⊗R⊗O	H

1.13.2 Dynamic (18 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Action	D⊗I⊗E	N	Fusion	S⊗D	D
Activation	A⊗D	D	Process	D⊗F⊗R	N
Alignment	I⊗A⊗S	N	Synergy	D⊗I⊗R	D
Amplification	Ft⊗D⊗It⊗R⊗O	D	Trajectory	A⊗D⊗F	N
Behavior	D⊗F⊗S⊗R	N	Transformation	S⊗I⊗D⊗F⊗V	N
Bifurcation	A⊗D⊗S	D	Trigger	D⊗I⊗V	N
Convergence	D	D	Tropism	A⊗S⊗D⊗F	N
Duplication	S⊗I⊗F⊗D⊗V	N	Event	D⊗I⊗O	N
FeedbackLoop	S⊗I⊗A⊗D⊗F⊗V⊗E	N	**Processor** †	S⊗I⊗F⊗D	N

1.13.3 Regulatory (8 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Balance	A⊗S⊗F⊗O⊗V	N	Regulation	A⊗S⊗F⊗V⊗R	N
Calibration	A⊗I⊗V	N	Scope	A⊗S⊗I	N
Constraint	S⊗I⊗F⊗V⊗R	N	Threshold	A⊗I⊗O	N
Homeostasis	A⊗S⊗F⊗V⊗E	N	Trade-off	A⊗I⊗F	D

1.13.4 Adaptive (7 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Adaptation	I⊗F⊗D	N	Resilience	A⊗S⊗E	N
Emergence	S⊗I⊗D⊗V	N	Self-Organization	A⊗I⊗D	N
Memory	D⊗F⊗D	N	Signature	S⊗I⊗A⊗O	N
Pattern	S⊗It⊗A⊗R⊗Im	N			

1.13.5 Energetic (5 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Dissipation	F⊗D	N	Storage	S⊗F⊗R	N
Entropy †	F⊗I⊗D	D	**Transducer †**	F⊗S⊗I	N
Gradient	F	N			

1.13.6 Informational (8 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Code	S⊗I⊗V	N	Representation	S⊗I⊗V⊗R	N
Coding	S⊗I⊗D⊗F⊗R	N	Signal	F⊗I⊗R⊗O	N
Language	S⊗I⊗F⊗A⊗R	N	ValueSpace	It⊗V⊗O⊗R⊗Im	N
Pattern	S⊗It⊗A⊗R⊗Im	N	Signature	S⊗I⊗A⊗O	N

1.13.7 Ontological (9 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Context	O⊗R⊗Im⊗E	N	State	I	N
Environment	F⊗I⊗O	N	Substrate	S⊗F⊗O	N
Gradient	F	N	System	S⊗F⊗V	N
Observer	A⊗I⊗R	N	Resource	F⊗I⊗R⊗Im	N
Space	S⊗I⊗D⊗E	N			

1.13.8 Relational (8 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Agent	S⊗I⊗D⊗A⊗E	N	Mediator	S⊗I⊗F⊗O	N
Hub	S⊗F⊗E	N	Node	S⊗I⊗F⊗Im	N
Identity	S⊗I⊗A⊗V⊗E	N	Relation	S⊗I⊗A⊗Im	N
Link	S⊗I⊗F⊗V	N	Role	S⊗I⊗F⊗O⊗R	N

1.13.9 Teleonomic (3 concepts)

Concept	Formula	Pol.	Concept	Formula	Pol.
Agent	S⊗I⊗D⊗A⊗E	N	Scope	A⊗I⊗S⊗V	N
Role	A⊗S⊗I⊗Im	N			

1.13.10 GenericConceptCombos (4 — defined in M2; instances in M1_CoreConcepts)

Combo	Formula	Components	Pol.
Inertia	S⊗F⊗I⊗D	⊗⇒(Memory, Entropy)	N
Potentialization	A⊗D⊗F	⊗⇒(Activation, Process)	D
Absorbing State	S⊗A⊗F⊗I⊗D	⊗⇒(Stase, Entropy)	N
Topological Defect	S⊗A⊗I⊗R⊗O	⊗⇒(Incoherence, Invariant)	N

1.14 Appendix D: Instance Catalog (TSCG v15.11.0)

All instances are hosted in the `instances/` directory of the TSCG repository and accessible via the Poclet Gallery: https://echopraxium.github.io/tscg/

1.14.1 D.1 Poclets (m3:Poclet) — see §4.2 for full table

22 validated poclets across 10+ domains. 10 are accompanied by standalone HTML simulations; the remaining 12 are in progress. Full details in the §4.2 table.

1.14.2 D.2 TransDisclets (m3:TransDisclet)

Systems existing natively across multiple disciplines simultaneously. No validated instances in v15.11.0. Candidates under analysis: the Fourier Transform (mathematics ↔ signal processing ↔ quantum mechanics) and the Predator-Prey dynamic (ecology ↔ economics ↔ epidemiology).

1.14.3 D.3 TSCG Tools (m3:TscgTool)

TscgPocletMiner — AI-powered corpus mining tool identifying poclet candidates from domain literature using RAG over the TSCG corpus. Implements a 5-round pipeline: candidate extraction → ASFID pre-screening → GenericConcept mapping → feasibility scoring → report. Built with ElectronJS + `@xenova/transformers`.

TscgPocletExplorer — Interactive browser for the poclet corpus. Provides filtering by domain, GenericConcept, ASFID coverage, and simulation type. Generates the `index.html` gallery via `generate_index.js`.

1.14.4 D.4 Systemic Frameworks (m3:SystemicFramework)

External systemic frameworks to be modeled within TSCG for comparison. Planned: VSM (Beer, 1972), TRIZ (Altshuller). No validated instances in v15.11.0.

1.14.5 D.5 Symbolic System Grammars (m3:SymbolicSystemGrammar)

I-Ching — The 64 hexagrams modeled as a symbolic generative grammar. Under analysis; not yet validated.

1.15 Appendix E: Framework Maturity Stages

Rather than presenting TSCG as a finished product, this appendix traces the progressive maturation of its key ideas. Each stage represents a genuine conceptual shift. Readers who disagree with a given stage's conclusions are encouraged to engage at precisely that level.

1.15.1 E.1 — Stage 1: Transversal Systemic Principles

The origin of TSCG is not a theory but an observation, accumulated over more than twenty-five years: *most systems share the same small set of structural behaviors*. A thermostat, a predator-prey ecosystem, a hormonal cascade, a blockchain consensus protocol, and a mythological cosmology all appear to regulate, cycle, amplify, and converge. The first maturity stage was the commitment to make this intuition *explicit and testable*: to enumerate candidate universal principles, name them, and ask whether they genuinely recur or merely appear to do so through loose analogy.

Key question: *Which principles are truly transversal, and which are domain-specific patterns in generic clothing?*

1.15.2 E.2 — Stage 2: The Observer Problem

Early attempts to enumerate systemic principles surfaced a fundamental ambiguity: the same system, observed at different scales or perspectives, yields different structural descriptions. A forest at the scale of individual trees exhibits competition; at ecosystem scale it exhibits homeostasis; at geological timescales it exhibits catastrophic bifurcation. This introduced the **Observer** as an irreducible component of any systemic model, with three irreducible parameters: *perspective*, *temporal scale*, and *spatial scale*. TSCG acknowledges the **anthropocentrism** inherent in a human-constructed vocabulary without claiming to transcend it.

Key question: *How do we build a generic vocabulary useful across observer perspectives without pretending to be perspective-free?*

1.15.3 E.3 — Stage 3: The Bicephalous Cyclops Metaphor

The Observer problem led to the **Bicephalous Cyclops** architecture: the Eagle Eye (ASFID) observing the Territory, the Sphinx Eye (REVOI) evaluating the Map. Each eye alone produces monocular, flat vision. Together they produce stereoscopic epistemic depth — the ability to measure not only the system, but the distance between our model of it and reality. This distance is the epistemic gap δ.

Key question: *How do we formalize the Map↔Territory feedback loop and make the gap between them quantifiable?*

1.15.4 E.4 — Stage 4: TSCG as a Meta-Map

With the bicephalous architecture established, TSCG needed epistemological positioning. It is not a theory of specific systems (that is what poclets are for). It is a **Meta-Map**: a vocabulary for constructing and comparing the Maps practitioners build of their Territories. The **echolocation metaphor** clarifies this: a bat does not build a visual representation of its environment — it emits signals and interprets echoes to construct a functional, navigation-adequate map. TSCG provides the signal vocabulary — GenericConcepts — with which practitioners construct navigation-adequate maps of their own systems.

Key question: *How do we ensure the Meta-Map remains a useful navigation tool rather than collapsing into self-referential abstraction?*

1.15.5 E.5 — Stage 5: The Poclet as Instrument

Abstract vocabulary untested against concrete systems is philosophy, not framework engineering. The **poclet** was developed as the primary instrument for four distinct, inseparable purposes: (1) **Popperian falsification** — each poclet is an opportunity to refute a claim; (2) **GenericConcept validation** — a concept earns M2 membership by demonstrating relevance across at least six unrelated domains; (3) **M2 purity preservation** — resisting the pressure to add domain-specific concepts to M2 is an active, ongoing discipline; (4) **simulation and pedagogical accessibility** — poclets are progressively accompanied by standalone HTML simulation accessible to anyone with a browser.

Key question: *How do we maintain M2 purity discipline while growing the poclet corpus rapidly enough to be credible?*

1.15.6 E.6 — Stage 6: Pedagogy, Democratization, and Research Potential

The final stage — ongoing, not concluded — is the explicit embrace of a pedagogical and democratizing mission alongside the research mission. The ability to recognize structural patterns across domains is a form of **systemic literacy** that equips practitioners, students, and citizens to reason about complex systems without disciplinary paralysis. The Systemic Esperanto acquires its full meaning here: not a replacement for domain vocabularies, but a shared medium of translation. The long-term scientific hypothesis: if cross-disciplinary systemic patterns are real, a sufficiently large poclet corpus analyzed with consistent tensor-product formalism should reveal **recurring structural signatures** cutting across domain boundaries. Whether TSCG's vocabulary is adequate to this task is an open question — submitted here, not answered.

Key question: *Can TSCG grow from a personal construction kit into a community instrument for systemic discovery?*

1.15.7 Summary Table

Stage	Core Shift	Key Concept Introduced
E.1	Intuition → Testable claim	Transversal systemic principles
E.2	System → System + Observer	Perspective, scale, anthropocentrism
E.3	Observer → Bicephalous architecture	Eagle Eye (ASFID), Sphinx Eye (REVOI), δ
E.4	Architecture → Epistemological positioning	Meta-Map, echolocation
E.5	Positioning → Empirical instrument	Poclet (test, discovery, purity, simulation)
E.6	Instrument → Community invitation	Systemic Esperanto, democratization

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End of Document

TSCG Framework v15.11.0 Authors: Echopraxium with the collaboration of Claude AI **Repository**: https://github.com/Echopraxium/tscg **License**: CC BY 4.0 (document) — BSD 3-Clause (source code) *Last Updated: March 2026*